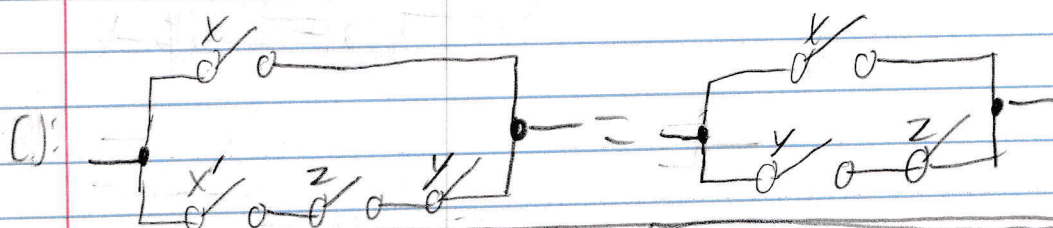
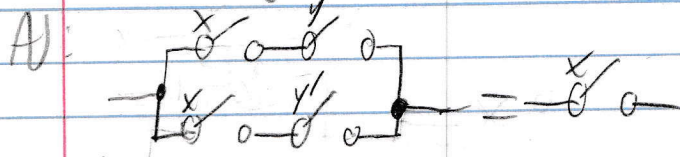


HW #2

2.10



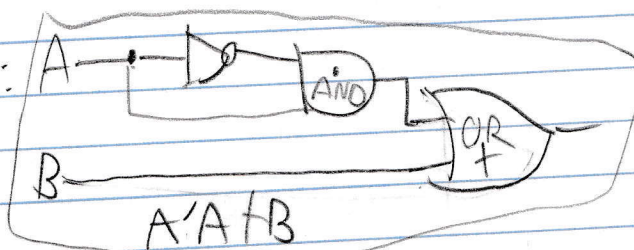
A) $(A' + B' + C)(A' + B' + C)' = [x \cdot x' = 0 \rightarrow \text{Laws of complementarity}]$

B) $AB(C' + D) + B(C' + D) = [xy + x = x = B(C' + D) \rightarrow \text{Absorption theorem}]$

C) $AB + (C' + D)(AB)' = [x + x'y = x + y = AB + C' + D \rightarrow \text{Elimination theorem}]$

2.13

A) output: $A'A + B + (B + B) \rightarrow \text{simplified: } A'A + B + 1 = A'A + B + 1 = A'A + B$

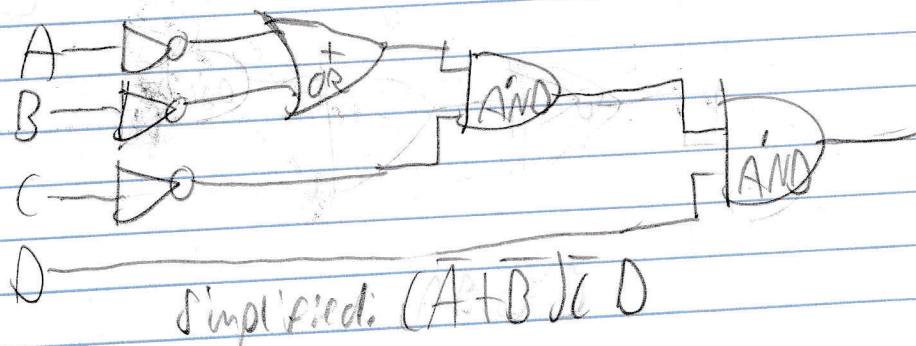


C) output: $(AB + C)'D \cdot (AB + C + D)$

$\overline{D}(x + D)$

$\overline{x}(D)$

$(\overline{A} + \overline{B})\overline{C}D$



2.15

$$A) [A + (\overline{B+C+D})]' [A + D + (BC' + BA)]'$$

$$[A + B + D + BC]$$

$$[A + B(C + \overline{C}) + D]$$

$$[A + B + \overline{C} + D] \cdot [A + B + D] \rightarrow ABCD + A\overline{B}D$$

2.16

$$A) [A \cdot (B + \overline{C+D})] + [A + D] \cdot B + (\overline{C+A})$$

2.24

$$A) [(x+y) + (x \cdot \overline{y})z] \rightarrow [x+y + x\overline{y}z] \rightarrow [x+y + xz]$$

$$\rightarrow \boxed{x+y+z}$$

$$C) [(AB) + (A+B+\overline{C}) + \overline{D}]' \rightarrow [(\overline{A+B})(\overline{A\overline{B}C})(\overline{C+D})]$$

$$[(\overline{A\overline{B}C})(\overline{C+D})] \rightarrow \overline{A\overline{B}C} + \overline{A\overline{B}C}\overline{D} = \boxed{\overline{A\overline{B}C}}$$

2.29

$$B) \begin{array}{c|c|c|c|c|c|c|c} x & y & z & xy & yz & \overline{x}+z & (x+y)(yz)(\overline{x}+z) & (x+y)(\overline{x}+z) \\ \hline 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{array}$$

equal